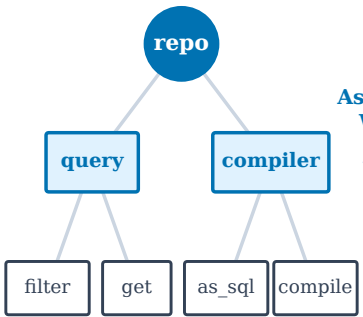
Codebase AST Graph

$\mathcal{G} = (\mathcal{V}, \mathcal{E}_{\text{call}} \cup \mathcal{E}_{\text{inherit}} \cup \mathcal{E}_{\text{import}} \cup \mathcal{E}_{\text{caller}})$

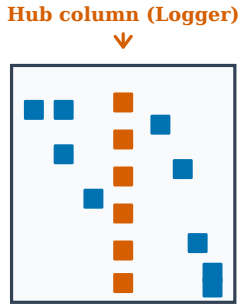


$|V| \approx 10^3 – 10^5$ *AST symbols*
Power-law: $D_{\text{hub}} \gg D_{\text{fn}}$

Static CSR Matrix

$W_{uv} = 0.50A^{(\text{call})} + 0.25A^{(\text{inherit})} + 0.15A^{(\text{import})} + 0.10A^{(\text{caller})}$

Asymmetric Weights



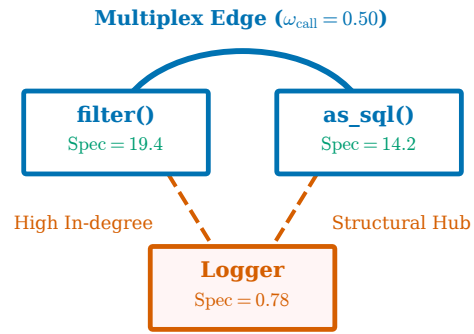
$T_{uv} = W_{uv}/D_u \quad d_u = I(D_u = 0)$

Zero-copy mmap binary arrays
< 1.0 ms ingestion overhead

Query-Directed Diffusion

$\text{Specificity}(v) = \frac{\pi_{\text{query}}(v)}{(\pi_{\text{global}}(v) + \epsilon)^{0.7}}$

Query Prior $\mathbf{p}_0$



Suppressed Hub (Spec < 1.0)

$\beta = 0.85$ *spectral convergence*
99.5% Hub Suppression Index

AST Slicing & Verification

Budget $K_{\text{effective}} = 3,480$ tokens $\leq 4,096$

Context Budget

```
class QuerySet:  # query.py:140-158
-   if key in clone.query:
+   parts = key.split('__')
+   if parts[0] in clone.query:

>> ast.parse() OK | Scope Validated
```

AST Breadcrumb Packing · **Lexical Scope Preserved**

Deterministic AST validation ($0.00 cost)
Bounded breadcrumb skeletons (<100 tokens/sym)